

Module 5: Concurrency and Synchronization in Operating Systems

Critical Section Problem: Mutual Exclusion, Progress, Bounded Waiting.

Semaphores: Counting and Binary Semaphores with wait and signal operations.

Classic Synchronization Problems: Bounded-Buffer, Readers-Writers, Dining Philosophers.

Deadlock Prevention: Negating one of four Coffman conditions.

Banker Algorithm for Deadlock Avoidance using safe state evaluation.